

Q: "What is the size of this room (in square meters)?"

(a) **Frame prompting:** no frame shows the room's extent — and more frames only add near-duplicates



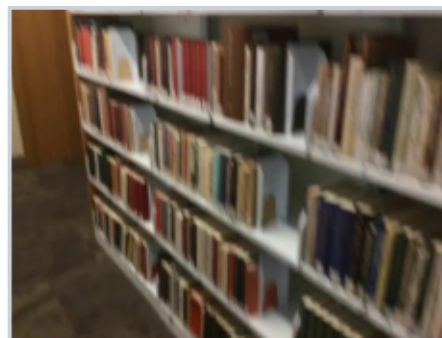
shelf close-up

camera
motion?
-----→



another shelf — or the same one?

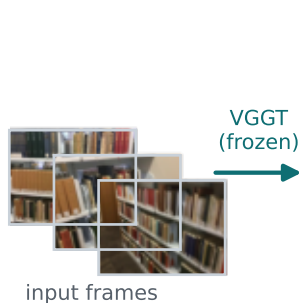
camera
motion?
-----→



position along the aisle unknown

all 8 input frames are close-ups taken while walking along the shelves; the aisle's length and the far wall are never visible, and the near-duplicate frames add no constraint
direct answer : 12.4 m² ✗ (GT 15.6)

(b) **ViewTree-D:** a depth-3 walk steps back and along the aisle until its extent is visible

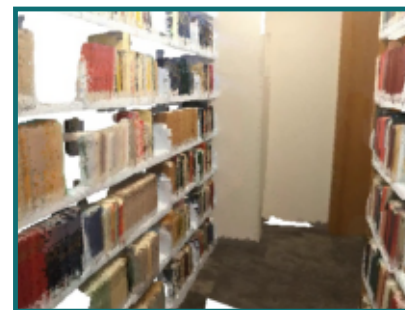


VGGT
(frozen)



explicit 3-D memory:
one unified reconstruction of the aisle

walk (human-constrained):
start at spot 2 →
FORWARD to spot 8 →
TURN RIGHT



acquired view: both shelf rows and the end wall in one frame — the extent is now visible

answer: 14.5 m² ✓
(GT 15.6; both retained walks agree, confidence above the direct guess)